

ACTIVITY 2 — LEARNER WORKSHEET**Lean, Six Sigma, Waste, Voice of Customer, and Value**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Activity	45 min	DEFINE	K1, A2

Name	
Date	

What you are doing

Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

What you will produce: A VOC-to-CTQ translation table with measurable targets, a value-added analysis and a waste walk log with every observation tagged to one of the eight wastes.

Tools and techniques: VOC, CTQ tree, DOWNTIME eight wastes, value-added analysis

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- voice-of-customer.csv — 10 rows (id, customer_statement, source, date)
- waste-walk-observations.csv — 10 rows (obs_id, observation, process_step, time_lost_min, waste_type)

templates/ — the worksheets you complete:

- value-added-analysis.csv (step_no, activity, VA_BVA_NVA, justification)
- voc-to-ctq.csv (voc_id, customer_statement, need, ctq_requirement, measure, target)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Tag every waste-walk observation with one of the eight DOWNTIME waste types.

Your steps

1. Capture the Voice of the Customer — record at least five verbatim customer statements.
2. Translate each VOC statement into a need, then into a measurable CTQ requirement.
3. Classify each process activity as value-added, business-value-added or non-value-added.
4. Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).
5. Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).
6. Identify which single waste type appears most often in your scenario.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

Step 1 — Capture the Voice of the Customer — record at least five verbatim customer statements.

Step 2 — Translate each VOC statement into a need, then into a measurable CTQ requirement.

Step 3 — Classify each process activity as value-added, business-value-added or non-value-added.

Step 4 — Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).

Step 5 — Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay...

Step 6 — Identify which single waste type appears most often in your scenario.

Answers to the discussion prompts

1. Which VOC statement was hardest to turn into something measurable, and why?

2. Where did you disagree about value-added versus business-value-added?

3. Which waste type appears most often — and what does that suggest about the cause?

Reflect & discuss

Which of the eight wastes is most visible in your own work this week?

What does your customer actually value that you have never measured?

Check your work

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

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